



UNIVERSITY OF KARACHI (UoK)
DEPARTMENT OF COMPUTER SCIENCE



COURSE INFORMATION SHEET

Session: 2026

Course Title: Calculus and Analytic Geometry

Course Code: BSCS-355 (1-B)

Credit Hours: 3+0

Semester: First

Pre-Requisite:

Instructor Name: Engr. Muhammad Ilyas

Email and Contact Information: 0332-3067025

WhatsApp Group:

Office Hours:

Mode of Teaching:

COURSE OBJECTIVE:

COURSE OUTLINE:



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COURSE LEARNING OUTCOMES (CLOs) and its mapping with GRADUATING ATTRIBUTES (GAs):

CLO No.	Course Learning Outcomes (CLOs)	GAs	Bloom's Taxonomy
1			
2			
3			

COMPLEX COMPUTING PROBLEM:

Complex Computing Problem Details	
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RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1 ()	CLO-2 ()	CLO-3 ()
Quizzes			
Assignments			
Midterm Exam			
Final Exam			



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GRADING POLICY:

Assessment Tools	Percentage
Quizzes & Assignments	20%
Midterm Exam	30%
Final Exam	50%
TOTAL	100%

Recommended Books:

Reference Books: Thomas Calculus



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LESSON PLAN

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Week No.	Week Dates	Topics	Required Reading	Key Dates
1	6-1-2026	Limits and Continuity: Introduction to functions, Introduction to limits technique of finding limits		
2	13-1-2026	Indeterminate forms of limits, Continuous and discontinuous functions and their applications.		
3	20-1-2026	Differential Calculus: Concept and idea of differentiation, Geometrical and physical meaning of derivative,		
4	27-1-2026	Rules of differentiation, Technique of differentiation,		
5	3-2-2026	Rate of change, Tangent and normal line, Chain rule.		
6	10-2-2026	Implicit differentiation, Linear approximation, Application of differentiation.		
7	17-2-2026	Extreme value function, Mean value theorem.		



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8	24-2-2026	Maxima and minima of a function for single variable, Concavity		
9	3-3-2026	Integral Calculus: Concept and idea of integration, Infinite integral		
10	10-3-2026	Technique of integration.		
11	17-3-2026	Riemann sums and definite integrals, Application of definite integral.		
12	24-3-2026	Improper integrals		
13	7-4-2026	Applications of integration: Area under the curve		
14	14-4-2026	Area under the curve Polar coordinate.		
15	21-4-2026	Analytic Geometry: Straight line in R^3 .		
16	28-4-2026	Equation for planes.		



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